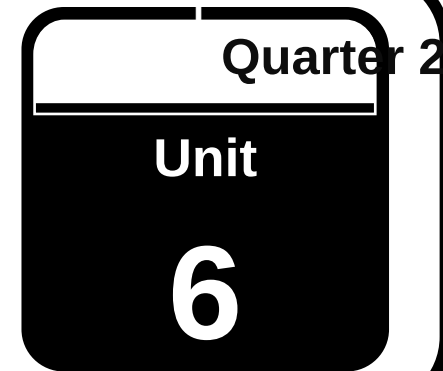


Senior High School



Lesson Exemplar in General Mathematics



GOVERNMENT PROPERTY
NOT FOR SALE

Lesson Exemplar for General Mathematics
Quarter 2: Unit 6

This material is intended exclusively for the use of senior high school teachers participating in the implementation of the Strengthened Senior High School Curriculum. It aims to assist in delivering the curriculum content, standards, and lesson competencies. Any unauthorized reproduction, distribution, modification, or utilization of this material beyond the designated scope is strictly prohibited and may result in appropriate legal actions and disciplinary measures.

Borrowed content included in this material are owned by their respective copyright holders. Every effort has been made to locate and obtain permission to use these materials from their respective copyright owners. The publisher and development team do not represent nor claim ownership over them.

Development Team

Writers: Lee C. Apas
Hernando C. Abalos, Jr.

Validator: Romel C. Anaya
Region 10

Language Editor: Ferdinand Kevin Gumalay

Consultant: Dr. Flordeliza F. Francisco
Far Eastern University

Learning Area Specialist: Wilson R. Santiago

Bureau of Learning Delivery
Bureau of Curriculum Development
Bureau of Learning Resources

Every care has been taken to ensure the accuracy of the information provided in this material. For inquiries or feedback, please write or call the Office of the Director of the Bureau of Learning Resources via telephone numbers (02) 8634-1072 and 8631-6922 or by email at blr.od@deped.gov.ph



Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

LESSON EXEMPLAR

Learning Area	GENERAL MATHEMATICS			Grade Level	11
Semester	FIRST	Quarter	SECOND		

I. OBJECTIVES

Content Standard	The learners demonstrate knowledge and understanding of statistical variables. <i>The learners demonstrate knowledge and understanding of the data collected involving school-based data, home or community data, and Online/Published Data.</i>
Performance Standard	By the end of the quarter, learners are able to interpret and explain how they make informed decisions based on the measures of central tendency and variability of a set of data collected for a specific purpose. <i>By the end of the quarter, learners are able to interpret and explain how they make informed decisions based on the measures of central tendency and variability of the data collected involving school-based data, home or community data, and Online/Published Data.</i>
Learning Competencies	The learners ... 1. identify different types of data. 2. classify a given set of data according to its level of measurement. 3. identify the appropriate measure of central tendency and measures of variability of a given data set for different purposes.

	4. use appropriate technologies (e.g. calculator, Excel, Jamovi, Geogebra, etc.) to calculate measures of central tendency and variability. 5. interpret the computed measures of central tendency and variability.
II. REFERENCES and MATERIALS	American Board for Certification of Teacher Excellence (n.d.). <i>Measures of Central Tendency and Variability</i>. https://www.americanboard.org/ptk/measures-of-central-tendency-and-variability/#4 Belecina, R. R., Baccay, E. S., and Mateo, E.B. (2016). <i>Statistics and Probability, 1st Edition</i>. Quezon City: Rex Book Store, Inc. pp. 21-45 Bluman, A. (2015). <i>Elementary Statistics: A Step-by-Step Approach</i>. London: McGraw-Hill Education - Europe. pp. 258-274 Brase, C. H. and Brase, C. P. (2004, 2019). <i>Understanding Basic Statistics 3rd and 8th editions</i>. Boston, USA: Cengage Learning. pp. 246-258 Bhandari, P. (2023, June 21). <i>Variability Calculating Range, IQR, Variance, Standard Deviation</i>. Scribbr. https://www.scribbr.com/statistics/variability/ Bhandari, P. (2023, June 21). <i>How to Calculate Variance Calculator, Analysis & Examples</i>. Scribbr. https://www.scribbr.com/statistics/variance/ Dela Cruz, E. O., et al. (2016). <i>Soaring 21st Century Mathematics: Statistics and Probability (K to 12)</i>. Quezon City: Phoenix Publishing House, Inc. pp. 16-25 Department of Education (2013). <i>Mathematics 8: Module 10 – Measures of Central Tendency and Measures of Variability</i>. DepEd. https://lrmds.deped.gov.ph/detail/4772 Illowsky, Barbara, and Susan Dean. 2018. <i>Introductory Statistics</i>. Houston, TX: OpenStax. https://openstax.org/books/introductory-statistics/pages/1-introduction. Laerd Statistics (2018). <i>Measures of Spread</i>. Laerd Statistics https://statistics.laerd.com/statistical-guides/measures-of-spread-range-quartiles.php Laerd Statistics (2018). <i>Absolute Deviation and Variance</i>. Laerd Statistics https://statistics.laerd.com/statistical-guides/measures-of-spread-absolute-deviation-variance.php Laerd Statistics (2018). <i>Standard Deviation</i>. Laerd Statistics

	https://statistics.laerd.com/statistical-guides/measures-of-spread-standard-deviation.php Pennsylvania State University (2025). <i>Measures of Variability</i> . Open.Ed@PSU. https://online.stat.psu.edu/stat500/lesson/1/1.5/1.5.3 Sternstein, M. (2007). <i>Barron’s AP Statistics 4th edition</i> . New York: Barron Educational Series. pp. 189 – 219 Triola, M. F. (2010). <i>Elementary Statistics Technology Update 11th edition</i> . USA: Pearson Education, Inc. pp. 208 – 218 Weather Atlas. (2024, February 7). <i>Climate and monthly weather forecast - Pag-asa, Philippines</i> . Weather Atlas https://www.weather-atlas.com/en/philippines/pag-asa-climate	
III. CONTENT	Data and Its Levels of Measurement <i>Recommended Time: 1 hour</i>	
IV. OBJECTIVES	At the end of the lesson, the learners should be able to: 1. identify types of data; 2. classify data according to its type; 3. identify levels of measurement; and 4. classify data according to its level of measurement.	
IV. PROCEDURES	ACTIVITIES (Instructions for Learners)	ANNOTATION (Instructions for Teachers)
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge Activity A.1: Data Around Us Instructions: List 5 pieces of information (data) you encounter or use every day — in school, at home/community, or online. Identify them whether they are numbers or words and determine if these can be ranked or measured.	Activity A.1: Data Around Us This activity is designed to activate learners’ prior knowledge by drawing from their real-life experiences with data before formally introducing the concepts of types of data and levels of measurement. To facilitate this activity, the teacher may use the following strategies. Strategy 1: Think-Pair-Share 1. Think - Give learners this prompt: List five 5 pieces of information (data) you encounter or use every day (e.g.in school, at home/community, or online).

		Examples learners might give: • My age • Number of steps I walked • Favorite color • Height of my classmates • Subjects I like most 2. Pair - Ask learners to pair - up and compare their lists. Allow them to discuss their outputs guided by the following questions: • Which of these are numbers? • Which are words or categories? • Which ones can you rank or measure? 3. Share - Draw a chart on the board with three (3) columns. <table border="1" data-bbox="1339 735 2119 1019"><thead><tr><th>Data that are Numbers</th><th>Data that are Words/Categories</th><th>Data that can be ranked or measured</th></tr></thead><tbody><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></tbody></table> Have learners volunteer examples from their lists and place them under the right column (they can guess at this point)	Data that are Numbers	Data that are Words/Categories	Data that can be ranked or measured															
Data that are Numbers	Data that are Words/Categories	Data that can be ranked or measured																		
	A.2. Establishing the Purpose of the Lesson Activity A.2: Why Data Matter? Instructions: Read the scenario and do the indicated tasks.	Activity A.2: Why Data Matter? To facilitate this activity, the teacher may apply the following strategy. Strategy 1: Scenario-Based Learning (SBL)																		

	Scenario: Imagine you are part of the Student Council, and your task is to create a survey to help improve the school canteen. You want to ask learners questions such as: • What food do you usually buy? • How much do you spend? • How often do you go to the canteen? • From 1 to 5, where 1 is the lowest and 5 is the highest, how would you rate the service. Tasks: In your own, list the different types of information (data) you need to gather from your schoolmates. Beside each piece of information, categorize them • whether it is a word or a number • whether it can be ranked or measured Answer why data matter in relation to your task?	Instructions for Teachers 1. Read aloud or display the canteen improvement scenario. Emphasize that learners must decide what kind of data to collect. 2. Instruct learners to list pieces of information they might collect for the survey and describe each as a word or number, and if it can be ranked or measured. 3. Have learners compare and discuss their answers with a partner, noting any differences in their reasoning. 4. Facilitate a short sharing session. Write a few examples on the board (e.g., favorite food = word, rating = number that can be ranked, amount spent = measurable number). Use probing questions like: - Is this data qualitative (word/categorical) or quantitative (number)? - Can this data be ordered or ranked? - Can this data be measured with a true zero? 5. After the sharing and discussion, guide learners with reflective questions such as: - Why do you think it's important to know whether data can be measured, counted, or ranked? - How can understanding data help you make better decisions or ask better questions? 6. Encourage learners to formulate their own conclusions. Then say: “Based on your ideas, how would you describe the purpose of today’s lesson?” 7. Write student responses on the board. Help them connect their thoughts to key ideas like: - Recognizing types of data
--	--	--

		- Understanding how data can be categorized - The importance of organizing data for real-world use 8. Let the class arrive at a collaborative summary statement, such as: <i>"We need to understand data types and how they're measured so we can collect the right information and use it wisely in situations like surveys, research, or decision-making."</i>
B. Instituting New Knowledge	B.1. Presenting Examples Activity B.1: How Worker Get Paid? Instructions: 1. Your teacher will show and explain real-life examples of different types of data and how they are measured. 2. For each example, the type of data (qualitative or quantitative) and the level of measurement (Nominal, Ordinal, Interval, or Ratio) are given. Also, explanations are provided. Example 1: Student Favorite Subjects Data: Math, Science, Filipino, English, Math • Type of Data: Qualitative • Level of Measurement: Nominal • Reason: These are categories with no order or numerical meaning. Example 2: Class Standing Based on Grades Data: 1st Honor, 2nd Honor, 3rd Honor, With Honors • Type of Data: Qualitative • Level of Measurement: Ordinal • Why? The data has a clear order but no exact measurable difference between the ranks.	Activity B.1: How Worker Get Paid? To facilitate this activity the teacher may apply the following strategy. Strategy 1: Socratic Questioning and Think-Pair-Share Instructions for teachers: 1. Display or read aloud each example (see Learners' Part) and ask guiding questions: • What kind of information do we have here? • Is this something we can count or measure? • Is there a meaningful order or difference? 2. Use Think-Pair-Share. Let learners first think individually. Then, have them pair up to compare their answers. Ask pairs or small groups to explain their reasoning to the class. 3. After all examples are discussed, ask: • What patterns do you notice between the examples?" • How would you explain each type of data and each level of measurement now?" 4. Record their ideas on the board to co-create a class summary. 5. Only step in to clarify or correct misconceptions as needed.

	Example 3: Daily Temperatures Data: 28°C, 30°C, 32°C, 27°C • Type of Data: Quantitative • Level of Measurement: Interval • Reason: The numbers have meaningful differences, but zero does not mean absence of temperature. Example 4: Student Heights Data: 150 cm, 160 cm, 155 cm, 170 cm • Type of Data: Quantitative • Level of Measurement: Ratio • Reason Heights are measurable, and zero height means no height; we can compare them using ratios. 3. Based on your observations, answer the following: • How do you know if data is qualitative or quantitative? • What makes data ordinal instead of nominal? • What makes data interval instead of ratio? 4. Try to think of other real-life examples for each type of data and level of measurement.	
	B.2. Discussing the Concept Activity B.2: Lesson Discussion: Getting to Know Me Instructions: Study the information and examples and prepare for class discussion. I. Types of Data In statistics, data refers to pieces of information collected for analysis. Understanding the type of data helps us choose the correct method for collecting, organizing, analyzing, and interpreting it. Data	Activity B.2: Lesson Discussion: Getting to Know Me To facilitate the discussion, teacher may select the following strategies: Strategy 1: Peer Teaching Teacher Instructions: 1. Create teaching stations for each type of data and level of measurement. 2. At each station, prepare: ○ The type of data and level of measurement with their respective definitions. ○ Examples in each type of data and level of measurement.

	generally falls into two main types: qualitative and quantitative. Qualitative Data (Categorical Data) This type of data describes qualities or characteristics. It cannot be measured numerically but can be grouped into categories. Definition: Data that represent attributes or labels, often using words or symbols. Examples: ○ Hair color: <i>black, brown, blonde</i> ○ Gender: <i>male, female</i> ○ Type of school: <i>public, private</i> ○ Marital status: <i>single, married, widowed</i> Quantitative Data (Numerical Data) This type of data deals with numbers and amounts. It can be counted or measured and supports mathematical operations. Definition: Data that represent numerical values. Examples: ○ Age: <i>16 years old</i> ○ Number of learners in class: <i>45</i> ○ Weight: <i>52.5 kg</i> ○ Income: <i>P25,000/month</i> II. Levels of Measurement Once we identify a variable as qualitative or quantitative, we also determine its level of measurement, which affects the type of analysis we can perform. There are four levels: nominal, ordinal, interval, and ratio, arranged from least to most informative.	3. Split the class into small groups and assign each group to a station. 4. Each group explains the type of data, level of measurement, and examples given at their station. 5. After the given time, rotate the groups to the next station. 6. When all groups have visited each station, bring the class together to discuss the learners understanding in each type of data and level of measurement.
--	--	--

Nominal Level (Categorical)

Definition: Data are labeled or categorized without any implied order or ranking.

Examples:

- Blood type: *A, B, AB, O*
- Type of fruit: *apple, banana, mango*
- Region: *NCR, CAR, Region IV-A*

Ordinal Level (Rank)

Definition: Data that can be ordered or ranked, but differences between ranks are not measurable.

Examples:

- Survey rating: *satisfied, neutral, dissatisfied*
- Class ranking: *1st place, 2nd place, 3rd place*
- Socioeconomic status: *low, middle, high*

Interval Level

Definition: Numerical data with measurable and meaningful differences between values, but no true zero; zero has meaning.

Examples:

- Temperature in Celsius: *30°C, 20°C*
- Dates: *2000, 2010*
- IQ scores: *90, 100, 110*

Ratio Level

Definition: Similar to interval level, but with absolute zero, zero means “none, nothing”.

Examples:

- Height: *160 cm*
- Weight: *60 kg*
- Age: *18 years*
- Income: *₱30,000/month*

B.3. Developing Mastery**Activity B.3: Scenario Worksheet****Instructions:**

Carefully, read each scenario provided on the table below. For every situation, determine whether the information presented is qualitative or quantitative and identify its corresponding level of measurement: nominal, ordinal, interval, or ratio. After identifying the type of data, explain your reasoning briefly in the last column.

Scenario	Type of Data	Level of Measurement	Reason
1. Ranking of top 5 learners based on academic performance			
2. Daily temperature recorded in Celsius for a week from PAGASA			
3. Number of hours spent on social media per day			
4. Preferred learning modality (modular, online, F2F, blended) of learners			
5. Amount of rice consumed by a household per week (in kg)			
6. The modes of transportation learners			

Activity B.3: Scenario Worksheet

In this activity learners will firm-up their learnings by answering scenario worksheet. Teacher may use the following strategy.

Think-Pair-Solve-Share

Teacher Instructions:

1. Give each student the scenario worksheet.
2. Ask them to answer the worksheet alone (Think).
3. Then, pair them with a seatmate to discuss the correctness of their answers together (Pair + Discuss).
4. Finally, invite volunteers or randomly chosen pairs to share their answers and explain their reasoning to the class (Share).
5. You may use visual aids like posters or digital slides to show the scenario worksheet.

	use to go to school (e.g., tricycle, jeep, walking, bicycle)				
	7. Monthly electricity consumption of your household in kilowatt-hours.				
	8. The type of music learners listen to while studying.				
	9. Number of correct items the student gets in a quiz.				
	10. Learners' satisfaction with school library using a 5-point scale with 1 as "Very Dissatisfied" to 5 as "Very Satisfied."				
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application				Activity C.1: Surveying Our School Life
	Activity C.1: Surveying Our School Life Instructions: Perform the given task. Scenario: - You will conduct a simple classroom or school-based survey to collect real data from your peers. Your goal is to apply what you've learned about types of data (qualitative or quantitative) and levels of measurement (nominal, ordinal, interval, ratio). - In small groups (at least 5 members), create 4–5 survey questions related to student life (e.g., study				
					Instructions: • Begin by reviewing sample survey questions with the class and discussing which types of data and measurement levels they represent. • Guide learners in forming realistic questions based on issues or experiences relevant to school or community life. • Allow learners to conduct surveys in a structured but independent way to build ownership of learning.

	habits, food preferences, daily screen time, number of subjects passed, satisfaction with school services). - Make sure each question represents a different type of data and level of measurement. - Survey at least ten 10 classmates and record the responses. - Organize your data in a table and identify: The type of data (Qualitative or Quantitative) and the level of measurement (Nominal, Ordinal, Interval, Ratio) - Reflect on how these data types affect how the information is analyzed or interpreted.	• After data collection, facilitate a group reflection through peer presentations or gallery walk where groups share their findings and discuss challenges in classifying data. • Ask guiding questions to deepen understanding.
	C.2. Making Generalization Activity C.2: Dear Data... Instructions: - Write a short creative letter addressed to “Data” as if it were a person. In your letter, express: • What you have learned about Data today. • Why Data is important in everyday life. • How you plan to apply what you have learned about types and levels of data. • Ask Data a question you’re still curious about. A Day in Data Instructions: - Think about your day today—from the time you woke up to now. List down at least 5 pieces of information (data) you encountered or generated today (e.g., number of hours you slept, your breakfast choice, your mood, the temperature, how many steps you took). For each, reflect on the following: • Is this qualitative or quantitative?	Activity C.2: Dear Data... For the summary best approach to be used is the Reflective Learning. To facilitate this part, you may select and facilitate one of the provided activities.

- What level of measurement does it belong to?
 - How could this data be useful in making decisions?
- Write your reflection in a paragraph form. Title it: “How Data Shapes My Day.”

C.3. Evaluating Learning

Activity C.3: Assessing Learning Progress

Instructions:

Carefully read each scenario in the provided table below. For each, identify the type of data (Qualitative or Quantitative) and the level of measurement (Nominal, Ordinal, Interval, or Ratio).

Scenario	Type of Data	Level of Measurement
1. Collecting the shoe sizes of volleyball players on the school team.		
2. Listing the favorite local fruits (e.g., mango, banana, lanzones) of market vendors.		
3. Recording monthly electricity bills (in pesos) from 10 households.		
4. Ranking the cleanliness of streets in the barangay from most to least clean.		
5. Measuring the time (in minutes) spent by learners traveling to school each day.		

Activity C.3: Assessing Learning Progress

This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation.

	6. Listing the types of pets owned by learners (e.g., dog, cat, bird, fish).									
	7. Recording the temperature (°C) inside classrooms throughout the day.									
	8. Counting how many plastic bottles are collected for recycling each week.									
	9. Categorizing different brands of detergents used in households.									
	10. Recording the ratings given by residents on barangay services from 1 (poor) to 5 (excellent).									
C.4. Additional Activities Activity C.4: Improving Understanding in Types of Data and Level of Measurement Remediation				Activity C.4: Improving Understanding in Types of Data and Level of Measurement This additional exercise may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning.						
	<table><tr><td>Scenario</td><td>Type of Data</td><td>Level of Measurement</td></tr><tr><td>1. The color of school uniforms (white, beige, blue).</td><td></td><td></td></tr></table>	Scenario	Type of Data	Level of Measurement	1. The color of school uniforms (white, beige, blue).					
Scenario	Type of Data	Level of Measurement								
1. The color of school uniforms (white, beige, blue).										

2. The number of siblings each student has.		
3. Ratings given to a barangay clean-up program (Very Poor to Excellent).		
4. Average height (in cm) of Grade 10 learners.		
5. Student ID numbers assigned to each student.		

Enrichment

Research Scenario

Title: *A Study on the Study Habits and Academic Performance of Senior High School Learners*

Statement of the Problem:

This study aims to determine the relationship between learners' study habits and their academic performance. Specifically, it seeks to answer the following questions:

1. What are the preferred study times (morning, afternoon, evening) of the learners?
2. How many hours do learners spend studying each day?
3. What are the learners' final grades in Mathematics for the last grading period?
4. What is the rank of learners based on their overall academic performance?

Task:

With this research, identify the possible data to be collected and classify them according to its type and level of measurement.

	<i>Note: The research questions provided are just part of the study. There could be other research questions needed to carry out the whole study.</i>												
III. CONTENT	Understanding Measures of Central tendency Recommended Time: 2 hours												
IV. OBJECTIVES	By the end of the lesson, learners will be able to: 1. identify the appropriate measure of central tendency of a given data set for different purposes; 2. use appropriate technologies (e.g. calculator, Excel, Jamovi, Geogebra, etc.) to calculate measures of central tendency; and 3. interpret the computed measures of central tendency.												
IV. PROCEDURES	ACTIVITIES	ANNOTATION											
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge Activity A.1: Getting Ready with 3Ms Instructions: Read the following scenarios and answer the questions that follow. Scenario 1 (First M): A learner helps his father by selling rice every after-school hour. As a curious student, he wants to know the average number of kilos he can sell in a week by recording his sales in kilos from Monday to Sunday as shown below. Based on the table below, what should be the average sale?	Activity A.1: Getting Ready with 3Ms For this activity you may use one of the following strategies. Rotating Stations Instructions for Teachers: 1. Prepares four (4) different stations, each focusing on one of the scenarios presented and one for the general questions: Station 1 for Mean (First M), Station 2 for Median (Second M), Station 3 for Mode (Third M), and station 4 for the general questions. 2. The class is divided into small groups of atleast 5 members, and each group is assigned to a different starting station (Excluding station 4). 3. Learners will read the scenario and discuss their reasoning as a group. 4. When the groups complete rotating through the three stations (Stations 1 to 3), they will then proceed to Station 4 and answer the follow-up questions.											
	<table><tr><th>Day</th><th>Sales (in Kilogram)</th></tr><tr><td>Monday</td><td>150kg</td></tr><tr><td>Tuesday</td><td>120kg</td></tr><tr><td>Wednesday</td><td>140kg</td></tr><tr><td>Thursday</td><td>120kg</td></tr><tr><td>Friday</td><td>110kg</td></tr></table>	Day	Sales (in Kilogram)	Monday	150kg	Tuesday	120kg	Wednesday	140kg	Thursday	120kg	Friday	110kg
Day	Sales (in Kilogram)												
Monday	150kg												
Tuesday	120kg												
Wednesday	140kg												
Thursday	120kg												
Friday	110kg												

Saturday	120kg
Sunday	140kg
Average Sale	

Scenario 2 (Second M): Making use of the sales of the student in the first situation, arrange the sales from least to greatest. What sale appears to be in the middle?

Day	Sales (in Kilogram)
Monday	150kg
Tuesday	120kg
Wednesday	140kg
Thursday	120kg
Friday	110kg
Saturday	120kg
Sunday	140kg
Average Sale	

Scenario 3 (Third M): Referring to the table of sales in Scenario 1, which sale occurred most often?

Day	Sales (in Kilogram)
Monday	150kg
Tuesday	120kg
Wednesday	140kg
Thursday	120kg
Friday	110kg
Saturday	120kg
Sunday	140kg
Average Sale	

Questions:

1. What is the measure of central tendency used in each scenario?
2. How are you going to determine the mean, median, and mode based on the three (3) scenarios?

5. Once all stations are done, the teacher gathers the class for a whole-group discussion, reviewing the correct answers and asking reflection questions.

Note: The teacher facilitates by rotating around the room, assisting and guiding learners as needed.

Think-Pair-Share

Instructions for Teachers:

1. The teacher begins by presenting the three situations either through a printed handout or projected on the board.
2. Learners are asked to first work individually (Think) to read the scenarios and answer the three guiding questions: What is the average (mean) sale in a week? What sale appears to be in the middle (median)? What sale occurred most often (mode)? They record their answers and computations in their notebooks.
3. Learners are then asked to pair up and share their answers with a partner. During this phase, they compare results, explain their reasoning, and help clarify any misunderstandings.
4. In the Share stage, several pairs are invited to present their answers to the class.
5. The teacher facilitates a discussion, summarizing each measure of central tendency, clarifying misconceptions, and asking reflective questions as a sort of reviewing concepts of measures of central tendency.

	3. What clue in each scenario helped you figure that out? 4. Can one data set have the same value for mean, median, and mode?	
	A.2. Establishing the Purpose of the Lesson Activity A.2: Student Quiz Scores: A Fair Representation Instructions; 1. Think about this situation and answer the questions that follow: - A teacher gave a 50-item quiz to his class. The following are the scores (out of 50) of 15 learners: 50, 49, 50, 47, 50, 45, 50, 42, 10, 50, 38, 48, 44, 50, 33 - At the end of the quiz week, the teacher wants to know the average score for the class, but he's unsure whether to use the mean, median, or mode to represent the learners' performance. Questions: 1. Which measure of central tendency (mean, median, or mode) do you think best represents the class's typical performance on the quiz? Why? 2. Discuss your reasoning with a seatmate. Did you choose the same measure? Why or why not?	Activity A.2: Student Quiz Scores: A Fair Representation To deliver this activity, you may apply the following strategy: Think-Pair-Share Instructions: 1. present the following real-life scenario to the learners either on the board, through a printed handout, or projected on screen. Say: <i>"Class, today we will start with a real scenario. Imagine that a teacher gave a 50-item quiz to 15 learners. These are their scores: 50, 49, 50, 47, 50, 45, 50, 42, 10, 50, 38, 48, 44, 50, and 33. Now, the teacher wants to report the average performance of the class, but he's unsure whether to use the mean, median, or mode to represent it. Your task is to figure out which of these three would give the most accurate and fair picture of how the class performed."</i> 2. Give learners time to silently reflect on the question by saying: <i>"Think about the scores carefully. Which measure of central tendency do you believe best represents the class's performance, and why? Consider how high or low scores might affect your choice."</i> 3. After a few minutes of quiet thinking, prompt learners to pair up and share their answers: <i>"Now, pair up with someone beside you and discuss your answer. Did you both choose the same measure? Talk</i>

		<i>about how you arrived at your conclusion. What makes one measure more appropriate than the others in this situation?”</i> 4. As they share, walk around and listen to their reasoning, offering guidance when necessary. After a few minutes of discussion, bring the class back together and invite pairs to share their insights. Use this opportunity to draw attention to key observations: <i>“Some of you may have noticed that the score of 10 is much lower than the rest—this kind of score is called an outlier. What happens if we use the mean when an outlier is present? Does it truly reflect how most learners did? What if we used the mode, which is 50 in this case? Is it enough to say most learners got 50, even if several didn’t?”</i> 5. As learners respond, steer the discussion toward recognizing that each measure tells a different story, and not all are appropriate in every situation. Emphasize: <i>“This situation shows us why it’s important not just to calculate the mean, median, or mode, but to think about which one is most appropriate based on the nature of the data. If we choose the wrong one, we might give a misleading picture of how learners actually performed. That’s exactly what today’s lesson is about—not just how to compute measures of central tendency, but when and why to use each one properly.”</i>
B. Instituting New Knowledge	<i>B.1. Presenting Examples</i> Activity B.1: Introducing Mean, Median, and Mode Instructions: 1. Read and understand the following examples and answer the questions provided.	Activity B.1: Introducing Mean, Median, and Mode To deliver this activity you may use any of the following approaches: Think–Pair–Share Instructions for Teachers:

	Scenario 1: A farmer harvested the number of mangoes from his trees for 7 consecutive days. The recorded harvests (in kilos) were: 20, 22, 24, 21, 23, 25, 22. Question: If the farmer wants to know his average daily harvest for the week to estimate how much he can supply regularly, which measure of central tendency should he use? Why? <i>Expected Answer:</i> The mean is appropriate here because the data is relatively close in value, and no extreme outliers exist. The farmer wants a reliable overall average, and the mean gives a fair estimate of his regular production. Scenario 2: A teacher gave a 50-item test. Here are the scores of her 15 learners: 50, 49, 50, 47, 50, 45, 50, 42, 10, 50, 38, 48, 44, 50, 33. Question: If the teacher wants to represent the typical performance of her class, which measure of central tendency should she use? Why? <i>Expected Answer:</i> The median is best in this case because the low score of 10 is an outlier that would significantly lower the mean. The median gives a better picture of a “typical” score in the presence of extreme values. Scenario 3: A shoe store recorded the shoe sizes of 30 customers who bought shoes in one day. The most frequently purchased size was size 8, which appeared 10 times, while other sizes appeared only 1–3 times each.	1. Begin by briefly reviewing the concepts of mean, median, and mode. Then, the three scenarios are presented one at a time, either via slides, printed handouts, or read aloud. 2. For each scenario, learners are first asked to think silently and independently about which measure of central tendency is most appropriate and why. 3. After a minute or two of individual thinking, the teacher instructs the class to pair up with a seatmate and discuss their reasoning. 4. Once pairs have had time to discuss, the teacher leads a whole-class sharing where several pairs are invited to share their conclusions. 5. The teacher ensures that the class identifies the reasoning behind each appropriate measure—e.g., the mean is used when data is balanced, the median when outliers exist, and the mode when frequency is key. Rotating Stations Instructions for Teachers: 1. Before class, the teacher sets up three stations around the classroom, each with one of the three scenarios: • Station 1: Mean scenario (Farmer’s mango harvest) • Station 2: Median scenario (Quiz scores with an outlier) • Station 3: Mode scenario (Shoe store sizes) 2. Each station includes the scenario, a guiding question, and space for learners to write or discuss their group response. 3. Learners are divided into small groups, at least 5 members, and assigned a starting station. 4. At each station, the group reads the scenario and answers the guiding question. 5. After all groups have visited all three stations, the class reconvenes for a plenary discussion.
--	--	--

	Question: If the store manager wants to decide which shoe size to restock more frequently based on customer demand, which measure of central tendency should be used? Why? <i>Expected Answer:</i> The mode is most appropriate because the manager is interested in the most common size purchased. The mode shows the most frequent data point, which is what matters in this business decision.	6. The teacher facilitates a class synthesis where each scenario is discussed, and groups share how they reasoned through the choice of mean, median, or mode. 7. The teacher emphasizes how context determines appropriateness, leading to a stronger conceptual understanding.
	C.2. Discussing the Concept Activity C.2: Understanding Each Measure of Central Tendency The discussion is divided into two (2) parts: Deep understanding of each measure of central tendency (With interpretation of the result), and Computation of Measures of Central tendency using Excel and Jamovi. Overview: Measures of central tendency are statistical tools used to describe the center or typical value of a set of data. They provide a single value that represents the entire distribution, allowing us to make generalizations, comparisons, and informed decisions based on data. The three most common measures are: • Mean – also called the average • Median – the middle value when data is ordered • Mode – the most frequently occurring value Each has specific uses depending on the nature and distribution of the data. Definition and Application of Each Measure <i>A. Mean (Average)</i>	Activity C.2: Understanding Each Measure of Central Tendency To facilitate the discussion of the following topics you may apply any the following strategies: Small Group Discussion Part A: Understanding Measures of Central Tendency Instructions: 1. Divide the class into small groups (4–5 learners per group). 2. Prepare printed or digital copies of the lesson handouts and sample dataset. 3. Distribute Part 1 of the handout to all groups and say: "Each group will now explore the concepts of mean, median, and mode—what they are, when to use them, and their strengths and limitations. You will also look at real-world examples and interpret the results." 4. Instruct groups to:

	<i>Definition:</i> The mean is the sum of all values divided by the number of values. <i>Application:</i> Used when data is uniform and free from extreme values (outliers). It is commonly used in computing averages like grades, temperatures, and performance scores. <i>Strengths:</i> • Uses all data values, providing a balanced summary. • Widely understood and accepted. <i>Weaknesses:</i> • Sensitive to outliers; extreme values can distort the result. • May not represent the typical value in a skewed distribution. <i>Example:</i> The quiz scores of 8 learners in a 10-item quiz are: 9, 8, 7, 10, 9, 8, 6, 10 Mean: 8.375 <i>Interpretation:</i> The mean score is 8.375, which suggests that, on average, learners performed well on the quiz. Since the scores are fairly close to each other and there are no extreme outliers, the mean effectively represents the central performance of the group. Most of the individual scores (6, 7, 8, 9, and 10) revolve around or are near the mean, indicating a consistent level of achievement among the learners. <i>B. Median</i>	• Read the definitions and examples in the handout. • Highlight key concepts and discuss their insights. • Answer guiding questions like: ◦ Which measure best represents the center if there are outliers? ◦ What does the result tell us about the overall data? 5. Circulate among groups. Prompt thinking with questions like: • “How would the mean change if we added an extremely high score?” • “Which measure would be best for analyzing income data?” Part B: Computing Measures Using Excel and Jamovi Instructions for Teachers 1. Distribute Part 2 of the handout and guide groups to open their laptops. 2. Say: “Now, you’ll learn how to compute mean, median, and mode using Excel and Jamovi. You’ll also explore which tool works better depending on the size of the data.” 3. Instruct groups to: • Enter or copy the sample dataset into Excel. • Follow the guide to compute: ◦ =AVERAGE() for mean ◦ =MEDIAN() for median ◦ =MODE.SNGL() for mode
--	---	---

	<i>Definition:</i> The median is the middle value of an ordered data set. If there's an even number of values, the median is the average of the two middle values. <i>Application:</i> Best used when data is skewed or has outliers, such as income distributions or test scores with very low or high extremes. <i>Strengths:</i> • Unaffected by outliers, making it a reliable measure in skewed data. • Useful when the exact value at the center is important. <i>Weaknesses:</i> • Does not use all values, so some information may be ignored. • Not suitable for nominal data. <i>Example:</i> Consider the scores of 9 learners: 45, 48, 50, 50, 50, 52, 55, 97, 100 Median: 50 <i>Interpretation:</i> Although one student scored 97 and the other one scored 100, the median score is 50, meaning that half of the learners scored below 50 and half scored above 50 and half scored above 50. The median gives a better representation of the overall performance because it is not affected by the two unusually high scores. <i>C. Mode</i>	• Copy and paste the same data into Jamovi. • Use the <i>Descriptives</i> module to compute the same measures. 4. Reporting and Class Discussion
--	---	--

Definition:

The mode is the value that appears most frequently in a data set.

Application:

Best used with categorical or nominal data (e.g., favorite color, shoe size) or when identifying the most common item is important.

Strengths:

- *The only measure that can be used with qualitative data.*
- *Easy to identify and interpret.*

Weaknesses:

- *A data set may have no mode, one mode, or multiple modes, which can complicate interpretation.*
- *Ignores most data values; may not reflect the center well.*

Example:

The shirt sizes sold in a store in one week:

M, L, M, XL, M, S, L, M

Step 1: Tally the frequencies:

- S = 1
- M = 4
- L = 2
- XL = 1

Mode = M

Interpretation:

The mode is M, indicating that medium-sized shirts were the most commonly sold. This is useful for inventory planning, as it reveals customer preference.

Computation of Measures of Central tendency using Excel and Jamovi

Measures of central tendency—mean, median, and mode—are essential tools in understanding and interpreting data. With the availability of modern statistical tools, computing these measures has become more efficient, accurate, and manageable, especially when dealing with large datasets. There are different appropriate technologies that can be used (e.g. calculator, Excel, Jamovi, Geogebra, etc.) and for this discussion, two commonly used tools in education and research, Microsoft Excel and Jamovi, will be introduced.

Using Excel

- Microsoft Excel is a powerful spreadsheet tool commonly used in data analysis due to its accessibility, simplicity, and built-in functions for statistical computations. It is especially useful for:

- Organizing raw data
- Calculating mean, median, and mode quickly using functions
- Generating visual aids (charts, tables, and graphs)
- Automating repetitive computations

Computation Process Using Excel

Assume your data is listed in column A (from A2 to A16).

A. Mean

- Step 1: Click on an empty cell where you want the result to appear.
- Step 2: Type the formula: =AVERAGE(A1:A15)
- Step 3: Press Enter. The cell will display the mean of the data.

B. Median

- Step 1: Click on another empty cell.
- Step 2: Type the formula: =MEDIAN(A1:A15)

- Step 3: Press Enter. The cell will display the median of the data.

C. Mode

- Step 1: Click on another empty cell.
- Step 2: Type the formula: =MODE.SNGL(A2:A16) (or =MODE.MULT(A2:A16) for multiple modes)
- Step 3: Press Enter. The cell will display the most frequently occurring value.

Note: If no value repeats, Excel will return a #N/A error for the mode.

Using Jamovi

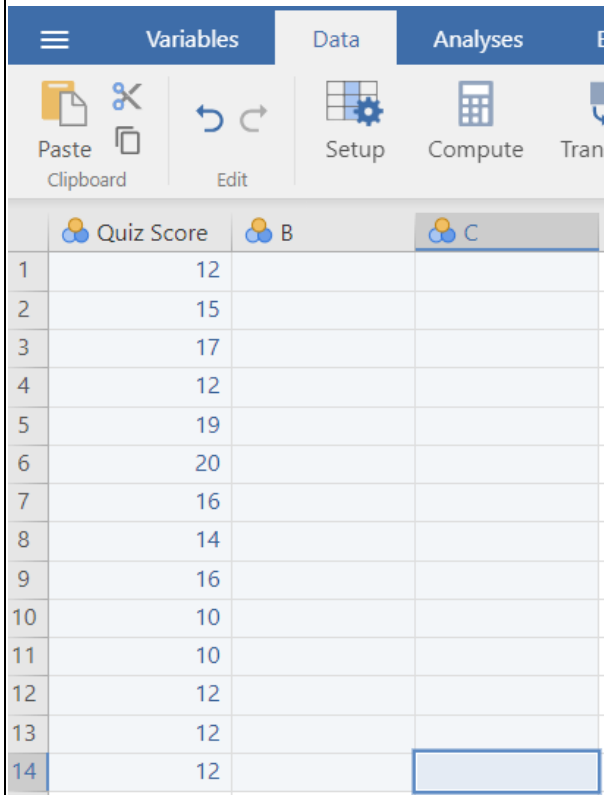
- Jamovi is a free and user-friendly statistical software designed for educators, learners, and researchers. Built on top of the statistical language, it combines the power of advanced analytics with an intuitive interface similar to spreadsheet applications. This makes it accessible even to beginners while being robust enough for professional research. It can be downloaded at <https://www.jamovi.org/download.html>

Step-by-Step: Computing Mean, Median, and Mode in Jamovi

Step 1: Prepare Your Data in Excel

Quiz Scores		
	12	
	15	
	17	
	12	
	19	
	20	
	16	
	14	
	16	
	10	
	10	
	12	
	12	
	12	

Step 2: Copy and Paste Data into Jamovi



The screenshot shows the Jamovi software interface. At the top, there are four tabs: 'Variables', 'Data' (selected), 'Analyses', and 'Export'. Below the tabs is a toolbar with icons for 'Paste' (Clipboard), 'Edit' (Undo/Redo), 'Setup' (Grid), 'Compute' (Calculator), and 'Transform' (Arrow). Below the toolbar is a data table with three columns: 'Quiz Score', 'B', and 'C'. The table has 14 rows, numbered 1 to 14. The 'Quiz Score' column contains the following values: 12, 15, 17, 12, 19, 20, 16, 14, 16, 10, 10, 12, 12, 12. The 'B' and 'C' columns are empty. The cell at row 14, column 'C' is highlighted with a blue border.

	Quiz Score	B	C
1	12		
2	15		
3	17		
4	12		
5	19		
6	20		
7	16		
8	14		
9	16		
10	10		
11	10		
12	12		
13	12		
14	12		

Step 3: Compute Mean, Median, and Mode

1. Click on the 'Analyses' tab at the top.
2. Under Exploration, click Descriptives.
3. Move the variable (Scores) to the right panel under Variables.
4. In the Statistics section, check the boxes for Mean, Median, and Mode.

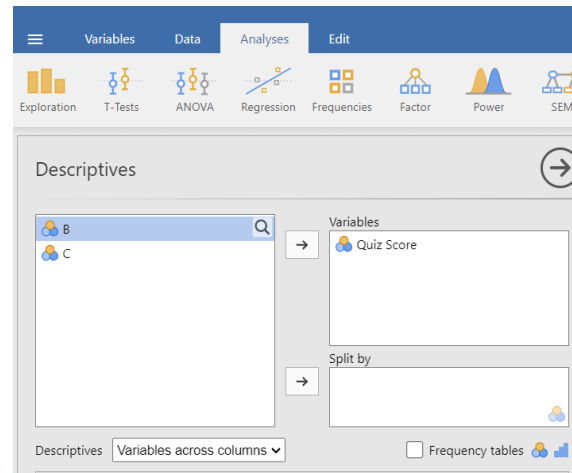
1.

	Quiz Score	B	C
1	12		
2	15		
3	17		
4	12		
5	19		
6	20		
7	16		
8	14		
9	16		
10	10		
11	10		
12	12		
13	12		
14	12		

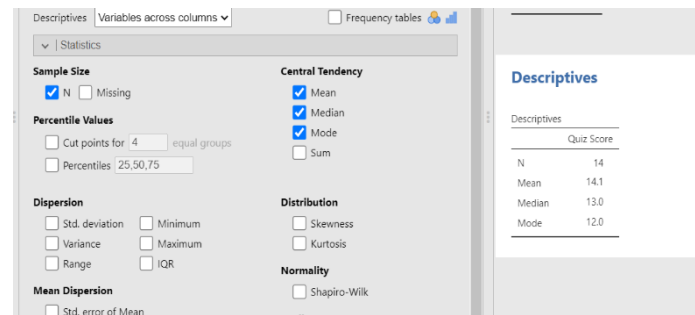
2.

	re	B	C
scatr	12		
	15		
Scatterplot	17		
Pareto Chart	12		
5	19		
6	20		
7	16		
8	14		
9	16		
10	10		
11	10		
12	12		
13	12		
14	12		
15			

3.



4.



Note: The results for mean, median, and mode are presented in the side panel.

B.3. Developing Mastery

Activity B.3: Practicing the Skills on Measures of Central Tendency

Instructions:

Read the following problems and answer what is asked.
Use Excel or Jamovi in the computation process.

1. A science teacher recorded the scores of 30 learners in a 100-point test. The scores were:

Scores:

86, 90, 88, 92, 89, 87, 85, 91, 90, 89,
88, 87, 86, 90, 88, 92, 89, 87, 88, 91,
85, 89, 90, 88, 87, 89, 91, 86, 88, 90

- a. Appropriate Measures of Central Tendency:
- b. Reason:
- c. Computed value:
- d. Interpretation:

2. A public school principal is analyzing the monthly family income (in PHP) of 31 learners to understand economic disparities:

Monthly Incomes (in PHP):

6,500; 7,000; 7,500; 8,000; 8,500; 9,000; 9,500;
10,000; 10,500; 11,000; 11,500; 12,000; 12,500; 13,000;
13,500; 14,000; 14,500; 15,000; 15,500; 16,000; 16,500;
17,000; 17,500; 18,000; 18,500; 19,000; 19,500; 20,000;
21,000; 23,000; 120,000

- a. Appropriate Measures of Central Tendency:
- b. Reason:
- c. Computed value:
- d. Interpretation:

3. A PE teacher records the shoe sizes of 40 learners:

Shoe Sizes:

7, 8, 8, 9, 8, 7, 8, 9, 10, 8,

Activity B.3: Practicing the Skills on Measures of Central Tendency

To facilitate this activity, the teacher may apply any of the following strategies.

Group-Based Data Analyst Simulation

Instructions:

1. Divide the Class: Assign each group a specific problem (Mean Group, Median Group, Mode Group).
2. Let each group answer the assigned problem using either Excel or Jamovi or other available technology.
3. Let each group analyze whether the measure is appropriate for the data.
4. Let each group interpret the results and prepare a short explanation.
5. Let the learners present their output.
6. Teacher facilitates a synthesis on when each measure is best used and how to interpret the result.

Think-Pair-Share with Whole-Class Reflection

1. Each student individually solves the three problems.
2. Learners pair up to compare answers and reasoning.
3. Select pairs to share their solutions and justifications.
4. Teacher discusses each problem to clarify things about measures of central tendency.

	7, 9, 8, 8, 9, 7, 8, 9, 10, 8, 8, 7, 8, 9, 8, 9, 10, 8, 7, 8, 8, 9, 7, 8, 9, 10, 8, 7, 8, 9 a. Appropriate Measures of Central Tendency: b. Reason: c. Computed value: d. Interpretation:	
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application Activity C.1: Choosing the Right Location for a New Community Health Center Instructions: Read the following scenario and answer the question that follows. Scenario: A local government unit is planning to build a new community health center to serve several barangays. To decide where it should be placed, they gathered data on the travel times (in minutes) of residents from each barangay to the proposed location. The travel times are as follows: Barangay A: 12, 15, 14, 16, 13, 60 Barangay B: 10, 11, 9, 12, 11, 50 Barangay C: 18, 20, 17, 19, 21, 75 The planning team wants to summarize this data to identify which location would result in the most reasonable average travel time for the majority of residents. Question: Which measure of central tendency should the planners use to make a fair decision? Why	Activity C.1: Choosing the Right Location for a New Community Health Center This activity is designed to be completed as group work. Ensure that all learners actively participate and collaborate effectively. This task is an opportunity for them to deepen their understanding on the appropriateness of measures of central tendency in a certain problem. It also develops learners understanding on how to interpret the value of every measure of central tendency.
	C.2. Making Generalization Activity C.2: Wrapping-up the Lesson	Activity C.2: Wrapping-up the Lesson

	“Dear Central Tendency...” Instructions: On a short bond paper or reflection sheet, write a creative letter addressed to <i>Central Tendency</i> as if it were a person. In your letter, include the following as guides: • What you learned about mean, median, and mode today. • Why these measures are important in understanding data. • A situation in your life where you think you could apply what you learned. • One question you still have about data or central tendency. One Thing I Learned, One I Can Apply Instructions: 1. On a small sheet or form, write down the following: ○ One thing you learned from today’s lesson (e.g., concepts of measures of central tendency). ○ One way you can apply it in real life (e.g., track my quiz scores and see how I’m improving). 2. Be specific and honest. Your answer will help your teacher understand what stood out to you.	Activity C.2: Wrapping-up the Lesson For the summary, you may select and facilitate one of the provided activities.
	C.3. Evaluating Learning Instructions: Read the following problems and answer what is asked. Use Excel or Jamovi in the computation process. 1. A call center supervisor recorded the number of minutes it took 30 customer service agents to resolve tickets in one day.	This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation.

	Response Times (in minutes): 12, 15, 13, 14, 16, 13, 12, 15, 14, 13, 16, 17, 13, 15, 14, 12, 15, 13, 14, 15, 16, 13, 14, 15, 14, 13, 16, 15, 14, 15 - Appropriate Measures of Central Tendency: - Reason: - Computed value: - Interpretation: 2. A teacher surveyed 31 learners about their weekly allowance (in PHP) to understand spending habits. Weekly Allowances (in PHP): 50, 60, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 210, 220, 1,000 - Appropriate Measures of Central Tendency: - Reason: - Computed value: - Interpretation: 3. An adviser recorded the number of times 40 learners raised their hands to answer questions during a week. Participation Counts: 2, 3, 3, 4, 3, 2, 3, 4, 5, 3, 2, 4, 3, 3, 4, 2, 3, 4, 5, 3, 3, 2, 3, 4, 3, 4, 5, 3, 2, 3, 3, 4, 2, 3, 4, 5, 3, 2, 3, 4 - Appropriate Measures of Central Tendency: - Reason: - Computed value: - Interpretation:	
	C.4. Additional Activities Activity C.4: Improving Understanding Remediation	Activity C.4: Improving Understanding This additional exercise may be given as an assignment or homework. Remediation should be provided to

	Instructions: Answer the following problems: 1. A restaurant manager recorded the total daily sales (in PHP) over 30 days to monitor business performance. Daily Sales (in PHP): 5,300; 5,500; 5,600; 5,800; 5,900; 6,000; 6,100; 6,300; 6,400; 6,500; 6,600; 6,700; 6,800; 7,000; 7,100; 7,300; 7,500; 7,600; 7,700; 7,800; 7,900; 8,000; 8,100; 8,300; 8,500; 8,600; 8,700; 8,800; 9,000; 25,000 a. Appropriate Measures of Central Tendency: b. Reason: c. Computed value: d. Interpretation: 2. A PE teacher tallied the number of sit-ups completed by 40 learners in one minute. Sit-Up Counts: 30, 32, 30, 28, 30, 29, 30, 31, 30, 29, 28, 30, 30, 32, 30, 30, 29, 28, 30, 30, 31, 30, 29, 30, 32, 30, 30, 28, 29, 30, 30, 30, 30, 29, 30, 31, 30, 28, 30, 30 a. Appropriate Measures of Central Tendency: b. Reason: c. Computed value: d. Interpretation: 3. A guidance counselor asked 31 learners how many hours of sleep they usually get each night. Sleep Hours: 4, 5, 5, 6, 6, 6, 6.5, 7, 7, 7.5, 8, 8, 8, 8, 8.5, 9, 9, 9, 9, 9.5, 10, 10, 10, 10.5, 11, 11.5, 12, 12, 12.5, 13, 14	learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning.
--	---	---

- a. Appropriate Measures of Central Tendency:
- b. Reason:
- c. Computed value:
- d. Interpretation:

Enrichment

Answer the following problem:

1. A gardener buys 10 packets of seeds from two different companies. Each pack contains 20 seeds and he records the number of plants which grow from each pack.

Company A	20	5	20	20	20	6	20	20	20	8
Company B	17	18	15	16	18	18	17	15	17	18

(a) Find the mean, median, and mode for each company's seeds.

(b) Which company has the higher mean?

(c) Which company do you think is better with regard to seeds production?

2. Given the data set: 2, 4, 4, 6, 8, what will happen to the mean, median, and mode if 3 is added to each value? How do they compare to the original?

3. Given the data set: 1, 3, 3, 5, 7, what will happen to the mean, median, and mode when 2 is multiplied to each value?

III. CONTENT

Understanding Measures of Variability

Recommended Time: 2 hours

IV. OBJECTIVES

At the end of the lesson, learners will be able to:

- 1. define and differentiate the different measures of variability (i.e. range, interquartile range, variance, and standard deviation)

	2. compute the different measures of variability that include solving real-life applications using open-source statistical software (e.g. MS Excel, Jamovi)											
	3. analyze and interpret the computed measures of variability											
IV. PROCEDURES	ACTIVITIES	ANNOTATION										
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge	Activity A.1: Comparing Spread										
	Activity A.1: Comparing Spread	For this activity, you may choose any of the following strategies.										
	Instructions:	Human Bingo Data Collection (Individual)										
	1. Collect data of first quarter examination. These show the different test scores of Grade 11 learners in General Mathematics.	Instructions for Teachers:										
	2. With your group, compare the spread and identify which data set have the largest and smallest spread.	1.Begin by preparing two 3x3 bingo cards containing different statements. Sample cards are given below.										
	3. Answer this question together: “Which data sets of score are more dispersed than the other? How do you know?”	Card 1 Find someone who...										
	4. Take turns sharing your group's answers. Be ready to explain:	<table><tr><td>loves to play any kind of sport</td><td>owns a pet</td><td>plays mobile legend</td></tr><tr><td>can play musical instrument</td><td>can sing the Philippine National Anthem</td><td>loves reading books</td></tr><tr><td>can speak more than one language</td><td>owns a mobile phone</td><td>is a SB19 fan</td></tr></table>		loves to play any kind of sport	owns a pet	plays mobile legend	can play musical instrument	can sing the Philippine National Anthem	loves reading books	can speak more than one language	owns a mobile phone	is a SB19 fan
loves to play any kind of sport	owns a pet	plays mobile legend										
can play musical instrument	can sing the Philippine National Anthem	loves reading books										
can speak more than one language	owns a mobile phone	is a SB19 fan										
	a. If two sections took the same test and have same mean score, can you conclude that they both performed well? Why or why not?	Card 2 Find someone who...										
		<table><tr><td>can sing the Philippine</td><td>loves watching Asianovela</td><td>is a Bini fan</td></tr></table>		can sing the Philippine	loves watching Asianovela	is a Bini fan						
can sing the Philippine	loves watching Asianovela	is a Bini fan										

<i>National Anthem</i>		
wants to pursue a career in Science or Mathematics	can recite the <i>Panunumpa sa Watawat ng Pilipinas</i>	is a member or officer of a student organization
uses AI tools	enjoys solving puzzles	can play musical instrument

2. Each learner must go around to find for his/her classmate who will match the statement in the card. Note: Learner must fill only one box in the card. No learner must fill two or more boxes even if the statement in the card is describing the same learner.

3. Instead of filling the boxes with their names, the learner will write the score he/she obtained in the first quarter examination in General Mathematics.

4. Once completed, the learner will compare now the spread of the two sets of data collected.

Scanning Code Data Collection (Group)

1. Begin by preparing 6 QR codes containing 3 different scores. The QR codes must be placed in the different parts or stations of the room.
2. Assign learners to be in a group. Each group must have 5 members.
3. Each group will scan 3 different QR codes for first data set and another 3 different QR codes for second data set. *Note: QR codes having the same set must be combined.*

3. Once completed, the learners will compare now the spread of the two sets of data collected.

A.2. Establishing the Purpose of the Lesson

Activity A.2: Predicting Variability

Instructions:

1. Think about this situation:

Every school year, the Department of Education conducts the baseline nutritional status assessment of the learners from all public elementary and secondary schools. Thus, body mass index was utilized as one indicator for the assessment. Body mass index is a measure derived from the weight and height of an individual. Suppose your teacher will assign you to get the height and weight of the two sets of learners. Each set is composed of 5 learners. Using a meterstick or a tape measure for height and weighing scale (mechanical or digital) for weight, you will list the data collected on a paper.

2. Discuss in pairs or small groups:

a. Which group has the largest spread in terms:

- (i) height; and
- (ii) weight?

b. Which group has the smallest spread in terms:

- (i) height; and
- (ii) weight?

c. What will happen with the spread of the data:

- (i) remove one data point?
- (ii) add one data point?

3. Why is it important to know how varied or dispersed the data are?

Activity A.2: Predicting Variability

To deliver this activity, you may apply any of the following strategies:

How Do We Measure Up?

Instruction for Teachers:

1. Group the learners with 4-6 members. Each group will have 2 learners (with codes A and B) to have their height and weight be measured.
2. Distribute the meta cards to be used in writing the height and weight measures of the two learners of each group. Sample card is given below.

Group Number	A Learners		B Learners	
	Height	Weight	Height	Weight
1				
2				
3				
4				
5				

3. Once the measuring height and weight of the considered learners from the group complete, they will compare the spread of the height and weight measures.

Traffic Light Variability

1. Prepare 3 cards (red, yellow, green) with datasets on the height (cm) and weight (kg) of learners. Sample datasets are given below.

Red card: H = {90, 92, 93, 95, 96}
W = {70, 72, 73, 75, 76}

		Yellow card: $H = \{80, 85, 90, 95, 100\}$ $W = \{70, 75, 80, 85, 90\}$ Green card: $H = \{75, 85, 90, 100, 120\}$ $W = \{65, 75, 80, 90, 100\}$ 2. Divide the class into 10 groups with at least 5 members. 3. Provide each group with the 3 cards with different numerical data. 4. Each group will now compare the spread of the data given to them.																								
B. Instituting New Knowledge	B.1. Presenting Examples Activity B.1: Measures of Variability into Context Instructions: 1. Read and understand the following situation: A student wants to track and compare 7-day temperature ($^{\circ}\text{C}$) from two different locations. The following data were obtained: <table><tr><th>Location</th><th>Day 1</th><th>Day 2</th><th>Day 3</th><th>Day 4</th><th>Day 5</th><th>Day 6</th><th>Day 7</th></tr><tr><td>A</td><td>30</td><td>32</td><td>35</td><td>35</td><td>34</td><td>32</td><td>33</td></tr><tr><td>B</td><td>37</td><td>35</td><td>36</td><td>37</td><td>34</td><td>37</td><td>35</td></tr></table> 2. Think of the following questions: a. Which location showed more variation in daily temperatures? Explain your answer. b. If this trend continues, which location is likely to experience more stable temperatures in the future? Why?	Location	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	A	30	32	35	35	34	32	33	B	37	35	36	37	34	37	35	Activity B.1: Measures of Variability into Context To deliver this activity, you may use any of the following approaches: 4Ts: Tracking Temperature Trends Together Instructions to Teachers: 1. Start by stating the rationale on analyzing the temperature trends or spread. Briefly introduce the datasets on the 7-day temperature from two different locations and what learners will be exploring. 2. Assign learners into pairs (preferably with seatmates). Provide them with the table and encourage them to visualize the data. Allow them to compare patterns across locations to find stability or variability. 3. Instead of computing the measures of variability, ask learners the following questions: <i>a. Which location showed more variation in daily temperatures? Explain your answer.</i>
Location	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7																			
A	30	32	35	35	34	32	33																			
B	37	35	36	37	34	37	35																			

	c. Based on the temperature data, which location might be more comfortable for outdoor activities? 3. Now imagine the following situations: a. What if the temperature suddenly dropped by 5°C in one location? How would that affect the trend? b. What if the highest recorded temperature was unexpectedly higher in the second week? How would this impact the spread of the data? c. What if the new temperature values were all close to the mean? Would the spread increase or decrease? 4. Reflect and discuss: a. What does a small spread tell us about a dataset compared to a large spread? b. How would adding more data points affect the distribution and spread of the dataset?	<i>b. If this trend continues, which location is likely to experience more stable temperatures in the future? Why?</i> <i>c. Based on the temperature data, which location might be more comfortable for outdoor activities?</i> 4. Use different imagining scenarios or what-if questions to guide learners to have deeper understanding. Encourage peer discussions by having learners justify their conclusions. <i>a. What if the temperature suddenly dropped by 5°C in one location? How would that affect the trend?</i> <i>b. What if the highest recorded temperature was unexpectedly higher in the second week? How would this impact the spread of the data?</i> <i>c. What if the new temperature values were all close to the mean? Would the spread increase or decrease?</i> 5. Wrap up with the reflection questions: <i>a. What does a small spread tell us about a dataset compared to a large spread?</i> <i>b. How would adding more data points affect the distribution and spread of the dataset?</i> CD (Climate Data Explorers) Instructions to Teachers: 1. Divide the class into small groups (4–5 learners per group). 2. Give each group the 7-day temperature data for Locations A and B. Allow them to spot patterns across locations to find which location has more changes or if there is a clear trend. 3. Instead of computing the measures of variability, let each group draw temperature trend lines based on the given data. Each group discusses answers to the following questions about temperature trends and variability. <i>a. Which location showed more variation in daily temperatures? Explain your answer.</i>
--	---	---

		<i>b. If this trend continues, which location is likely to experience more stable temperatures in the future? Why?</i> <i>c. Based on the temperature data, which location might be more comfortable for outdoor activities?</i> 4. Each group presents their answers to the class. Compare insights between different groups. 5. Ask the following follow-up questions to help learners develop deeper understanding and explore the concepts further. <i>a. What if the temperature suddenly dropped by 5°C in one location? How would that affect the trend?</i> <i>b. What if the highest recorded temperature was unexpectedly higher in the second week? How would this impact the spread of the data?</i> <i>c. What if the new temperature values were all close to the mean? Would the spread increase or decrease?</i> 6. Wrap up with the reflection questions: <i>a. What does a small spread tell us about a dataset compared to a large spread?</i> <i>b. How would adding more data points affect the distribution and spread of the dataset?</i>
	B.2. Discussing the Concept Activity B.2: Understanding Measures of Variability Discussing the concept is divided into 2 parts: (i) deep understanding of each measure of variability (definition, formula, real-life application) and (ii) computation of measures of variability using open-source statistical software (e.g. MS Excel and Jamovi) 1. Understanding Measures of Variability <i>a. Introduction to Measures of Variability</i> Variability describes the spread or dispersion of the data (i.e. the distance between data points in relation to one another and to the mean of the distribution). In addition to measures of central tendency (i.e. mean,	Activity B.2: Understanding Measures of Variability To facilitate the discussion of the following topics, you may apply any the following strategies: Small Group Discussion Part A: Understanding Measures of Variability Instructions for Teachers 1. Divide the class into small groups (4–5 learners per group). 2. Prepare printed or digital copies of the lesson handouts and sample dataset.

	media, mode), measures of variability provide descriptive statistics that summarize the dataset. While measures of central tendency describe the typical value, measures of variability show how much the data fluctuates around that central value. 2. Measures of Variability a. <i>Range (R)</i> Range tells the spread of the data that involves the lowest and highest value. It is the simplest measure of variability by calculating the difference between the highest and lowest values in a dataset. It is particularly helpful in situations where understanding extreme values is important. Formula: $R = \text{Highest Value (HV)} - \text{Lowest Value (LV)}$ Example 1: In a certain test in General Mathematics, the 40 Grade 11 learners scored from 14 to 39. Hence, the range is 25 (i.e. $R = 39 - 14 = 25$). Example 2: Given the following data on the number of minutes to solve a math problem: {23, 10, 18, 19, 43, 16}. The range is 33 (i.e. $R = 43 - 10 = 33$). Note: While range is easy to compute, it does not account for how values are distributed across a dataset. b. <i>Interquartile Range (IQR)</i> The interquartile range represents the spread of the middle 50% of the data. It is obtained by computing the difference between the third quartile (Q_3) and first quartile (Q_1) or between the 75th percentile and 25th percentile. It is useful when we find to measure the variability while minimizing the impact of the extreme values.	3. Distribute Part 1 of the handout to all groups and say: <i>"After exploring different measures of central tendency, each group will now explore the concept of measures of variability (i.e. range, interquartile range, variance, standard deviation). You will also look at real-world scenarios and interpret the results."</i> 4. Instruct each group to: • Read the definitions and examples in the handout. • Highlight key concepts and discuss their insights. • Answer guiding questions like: ◦ <i>What does variability mean in a dataset?</i> ◦ <i>Why is it important to measure variability in analyzing the data on a given situation?</i> ◦ <i>How does variability help in comparing different datasets?</i> 5. Circulate among groups. Prompt thinking with questions like: • <i>In what scenarios would you use each measure?</i> • <i>What happens to measures of variability if we add or multiply a constant to each observation in a data set?</i> • <i>What does zero variability indicate about the data distribution?</i> • <i>Can you give an example of a data set with zero variability?</i>
--	---	--

Formula: $IQR = \text{Third quartile } (Q_3) - \text{First quartile } (Q_1)$
 $= 75^{\text{th}} \text{ percentile} - 25^{\text{th}} \text{ percentile}$

Example 3: Given the following data on the test scores of 10 Grade 11 learners in General Mathematics: {35, 32, 37, 45, 15, 23, 26, 26, 39, 40}. Thus, the **interquartile range (IQR) is 12.50**. *This means that the middle 50% of learners' test scores in General Mathematics are spread across 12.50 points. While test scores do vary, they are not extremely scattered.*

Note: Since IQR is not affected by the extreme values, it is a resistant measure of variability. It tells us how spread out the central portion of the data is, without being affected by the outliers.

c. Variance

The variance represents the mean of squared differences from the average. It measures the extent to which a set of data is dispersed from the mean value. There are two types of variance: *sample variance* (s^2) and *population variance* (σ^2). It is useful in analyzing how spread out the data is from the mean.

Formula:

$$\text{Sample variance: } s^2 = \frac{\sum (X - \bar{X})^2}{n-1} \quad \text{or} \quad s^2 = \frac{\sum f(X - \bar{X})^2}{n-1}$$

where n is the sample size and $\bar{X}$ is the sample mean.

Part B: Computing Measures of Variability Using Open-Source Statistical Software (e.g. Excel and Jamovi)

Instructions for Teachers

1. Distribute Part 2 of the handout and guide groups to open their laptops.

2. Say:

“Now, you’ll learn how to compute range, interquartile range, variance, and standard deviation using open-source statistical software like Excel and Jamovi. You’ll also explore which tool works better depending on the size of the data.”

3. Instruct groups to:

- Enter or copy the sample dataset into Excel.
- Follow the guide to compute:
 - =MAX()-MIN() for range
 - =QUARTILE(array,3)-QUARTILE(array,1) for interquartile range
 - =VAR.S() for variance
 - =STDEV.S() for standard deviation
- Copy and paste the same data into Jamovi.
- Use the *Descriptives* module to compute the same measures.

4. Reporting and Class Discussion

Population variance: $\sigma^2 = \frac{\sum (X-\mu)^2}{N}$ or $\sigma^2 = \frac{\sum f(X-\mu)^2}{N}$

where μ is the population mean, and the summation is over all possible values of the population and N is the population size.

Example 4: Given the same data on the test scores of 10 Grade 11 learners in General Mathematics: {35, 32, 37, 45, 15, 23, 26, 26, 39, 40}, the **variance is 84.18**. *This indicates that the test scores of the 10 Grade 11 learners have a moderate to high spread from the mean. Some learners performed better or lower than others.*

d. *Standard Deviation*

The standard deviation is a measure of variability that indicates how scattered data values are around the mean. For a given situation, it tells us whether the data points are close together or widely spread apart. It is computed by getting the square root of the variance.

Formula:

Sample standard deviation: $s = \sqrt{s^2} = \sqrt{\frac{\sum (X-\bar{X})^2}{n-1}}$ or

$$s = \sqrt{s^2} = \sqrt{\frac{\sum f(X-\bar{X})^2}{n-1}}$$

Population standard deviation: $\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum (X-\mu)^2}{N}}$
or

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum f(X-\mu)^2}{N}}$$

Example 5: Using the same data on the test scores of 10 Grade 11 learners in General Mathematics in Example 3: {35, 32, 37, 45, 15, 23, 26, 26, 39, 40}, the **standard deviation is 9.17**. *This indicates that the performance of learners in General Mathematics vary moderately from the average score. Test scores are somewhat scattered, meaning learners didn't perform identically. Some learners scored higher or lower than others.*

Note: If data points are far from the mean, then the standard deviation is high, meaning there is a greater variability. On the other hand, if data points are close to the mean, then the standard deviation is low, meaning there is more consistency on the data values given a situation.

Computation Process Using Open-Source Statistical Software Tools

A. MS Excel

Suppose there are 30 data points on a certain situation. Input the data in column A and rows 2 to 31 (i.e. cells A2 to A31). Cell A1 (column A, row 1) will be used for the name of the data.

1. Range

Step 1: Click on an empty cell where you want the result to appear.

Step 2: Type the formula: =MAX(A2:A31)-MIN(A2:A31)

Step 3: Press Enter. The cell will display the range of the data.

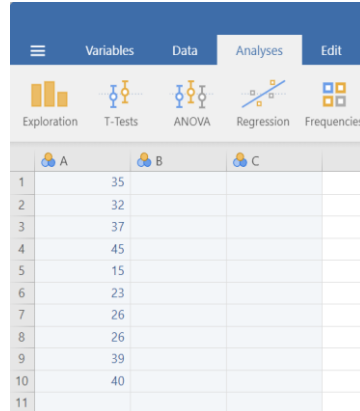
2. Interquartile Range

	Step 1: Click on another empty cell where you want the IQR result to appear. Step 2: Type the formula: =QUARTILE(A2:A31,3)-QUARTILE(A2:A31,1) Step 3: Press Enter. The cell will display the IQR result of the data. 3. Variance Step 1: Click on another empty cell where you want the variance result to appear. Step 2: Type the formula: =VAR.S(A2:A31) Step 3: Press Enter. The cell will display the variance result of the data. 4. Standard Deviation Step 1: Click on another empty cell where you want the standard deviation result to appear. Step 2: Type the formula: =STDEV.S(A2:A31) Step 3: Press Enter. The cell will display the standard deviation result of the data. Note: In most statistical software and calculator functions, <i>sample variance and sample standard deviation</i> are the default calculations. B. Jamovi Given the following data on the test scores of 10 Grade 11 learners in General Mathematics: {35, 32, 37, 45, 15, 23, 26, 26, 39, 40}. Step 1. Prepare/input the data in MS Excel.	
--	---	--

	A
1	Test Scores
2	35
3	32
4	37
5	45
6	15
7	23
8	26
9	26
10	39
11	40

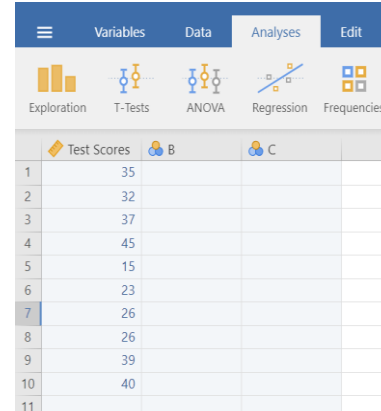
Step 2. Select the data on test scores (e.g. cells A2:A11) and paste it into Jamovi.

To rename a variable, double-click the column header (variable name) in the Data tab. This opens the "Variable Editor" where you can change the variable name, data type, and other properties. In this example, variable A renamed to **Test Scores**, measure type: **continuous**.



This screenshot shows the Jamovi interface with the 'Data' tab selected. The variable 'A' is highlighted in the column header row. The data table below shows the following values:

	A	B	C
1	35		
2	32		
3	37		
4	45		
5	15		
6	23		
7	26		
8	26		
9	39		
10	40		
11			

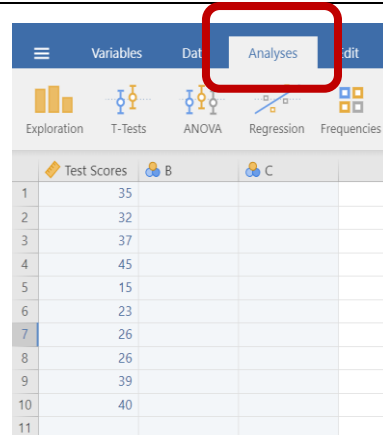


This screenshot shows the same Jamovi interface after renaming variable A to 'Test Scores'. The data table is identical to the previous one:

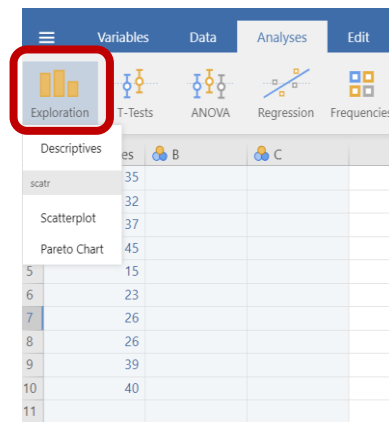
	Test Scores	B	C
1	35		
2	32		
3	37		
4	45		
5	15		
6	23		
7	26		
8	26		
9	39		
10	40		
11			

Step 3: Compute for the range, interquartile range, variance, and standard deviation.

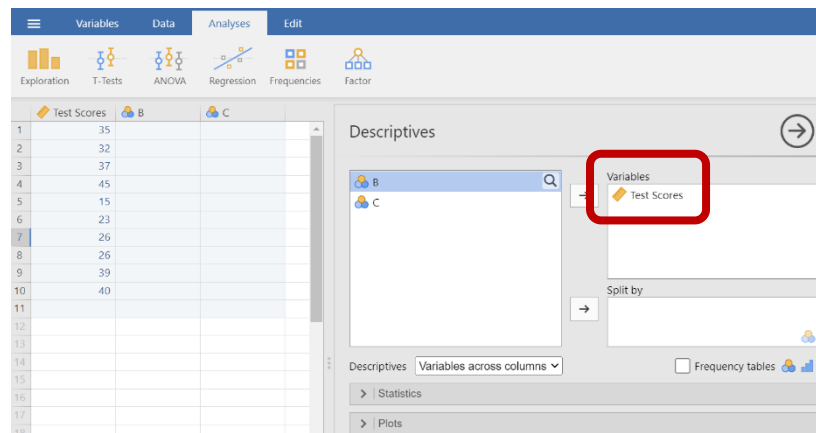
a. Click on the Analyses tab at the top.



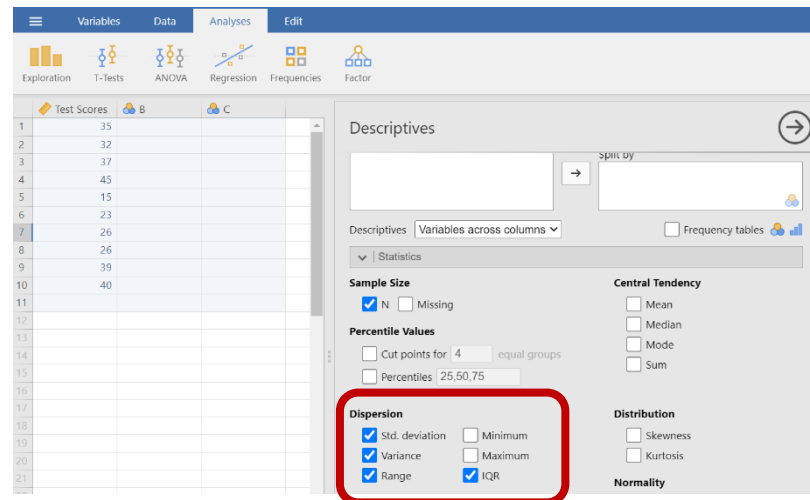
b. Under Exploration tab, click Descriptives.



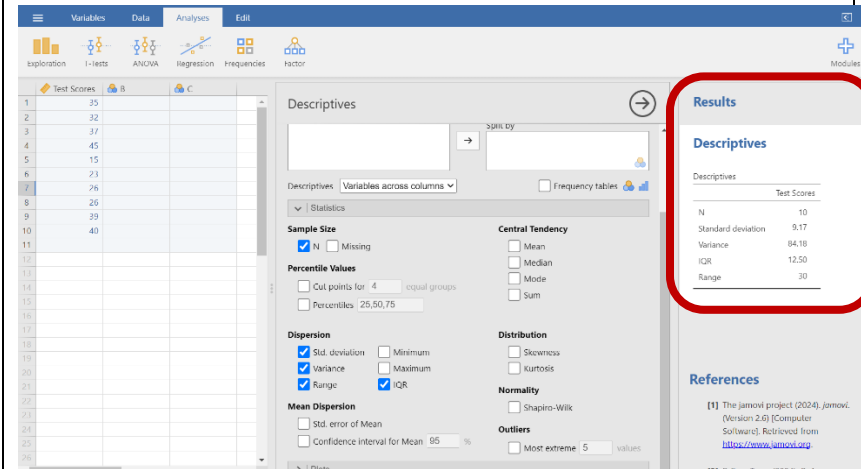
c. Move the variable (Test Scores) to the right panel under variables.



d. In the Statistics section, check the boxes for Range, IQR, Variance, and Standard Deviation.



e. The computed range, interquartile range, variance, and standard deviation of the given data set will automatically display in the side panel.



B.3. Developing Mastery

Practicing the Skills

Instructions for Learners:
Read and solve each problem.

1. Scenario:

A group of Grade 11 learners receives a weekly allowance (in pesos) from their parents. They collect data from 35 Grade 11 learners over one week:

700	850	750	1000	1150
450	500	1100	1200	1250
1300	650	950	1000	1500
2000	450	850	650	550
1250	525	800	750	650
1250	450	500	650	1350

Task:

a. Calculate the range, interquartile range, variance, and standard deviation.

To facilitate this activity, the teacher may apply any of the following strategies.

Variability Group Collaboration

Instruction for Teachers

1. Divide the class into four groups. Assign each group with one scenario involving measures of variability. Ensure each group has access to available technology tools (e.g. MS Excel, Jamovi) to compute and visualize data.
2. Each group inputs the data in the available technology tools to compute for the measures of variability.
3. Each group analyzes and interprets the results. Discuss insights within the group before presenting the findings.
4. Each group presents their output to the class. Encourage questions and feedback from the class
5. Teacher facilitates a synthesis on when each measure is best used and how to interpret the result. Summarize key takeaways for practical application of measures of variability.

- b. What can you say about the variance and standard deviation?
- c. Which of the measures of variability is considered reliable for the scenario? Explain your answer.
- d. What happens to the measures of variability if we add or multiply a constant to each observation in a data set?

2. Scenario:

Your school notices that learners struggle with concentration during peak heat index hours. Teachers track heat index variability to determine whether afternoon classes should be adjusted. Over the 30-day observation, the following data were recorded.

23	33	39	36	34	35
25	32	34	37	35	36
26	30	35	34	32	35
27	31	32	29	31	34
28	37	33	30	33	37

Task:

- a. Find the range of the dataset.
- b. Compute the interquartile range.
- c. Compute the variance and standard deviation.
- d. Why is it important to measure the variability of the heat index rather than just average values?
- e. What happens to the measures of variability if we add or multiply a constant to each observation in a data set?

3. Scenario:

A teacher wants to assess the consistency of learner performance in a General Mathematics exam. He/she collects scores from a class of 50 learners and groups them into intervals. The data are given below.

Technology-driven Problem Solving

Instruction for Teachers

1. Learners select or are assigned one of the four problem sets.
2. Learners solve the problem using available technology tools (e.g., MS Excel, or Jamovi) computing for the given measures of variability.
3. While working or having independent exploration, ask questions or offer feedback to deepen their reasoning skills.
4. Learners analyze and interpret the result. Explain what the variability measures revealed by the data.
5. Compare results between the four problem sets and discuss real-world applications which variability measure was most useful in each scenario.

41	43	22	21	18
23	35	34	32	33
24	14	15	23	32
37	37	36	34	35
33	32	21	18	19
20	21	22	23	21
21	23	24	25	27
18	19	29	19	30

Task:

- Find the range, interquartile range, variance, and standard deviation.
- Is range the most appropriate measures of variability for the given data? Why or why not? How about the interquartile range? How about the variance? How about the standard deviation? Explain your answer.
- What happens to the measures of variability if we add or multiply a constant to each observation in a data set?

4. Scenario:

A teacher records the daily screen time (in hours) for 30 learners over a week. The learners' screen time varies based on study habits, gaming, and social media use. The data are given below.

5	6	3	9	5	7
4	10	8	6	6.5	7.5
8.5	9	7.5	6.5	5.5	4.5
9	8	7.5	7.5	8	10
9	3	4	4.5	5.5	10.5

Task:

- Determine the range, interquartile range, variance and standard deviation.
- How do measures of variability in screen time affect focus and productivity?

	c. Which measures of variability is the most appropriate for the given data? d. What happens to the measures of variability if we add or multiply a constant to each observation in a data set?																																																			
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application Activity C.1: Watts Up? Investigating the Variations in Household Electricity Bills Problem: A certain electric cooperative wants to determine and analyze the variability in household electricity bills (in pesos) in one town to identify patterns in energy consumption. The collected data on the monthly electricity bills of 50 households are given below: <table><tr><td>1550</td><td>1450</td><td>1350</td><td>1500</td><td>1800</td></tr><tr><td>750</td><td>1550</td><td>1200</td><td>1300</td><td>1200</td></tr><tr><td>2500</td><td>1500</td><td>1600</td><td>1700</td><td>1500</td></tr><tr><td>2500</td><td>3000</td><td>3500</td><td>4000</td><td>3500</td></tr><tr><td>2300</td><td>3400</td><td>2300</td><td>2450</td><td>2000</td></tr><tr><td>2000</td><td>2100</td><td>1250</td><td>1300</td><td>1100</td></tr><tr><td>950</td><td>850</td><td>800</td><td>900</td><td>950</td></tr><tr><td>1250</td><td>1300</td><td>1200</td><td>1500</td><td>850</td></tr><tr><td>875</td><td>900</td><td>1350</td><td>750</td><td>780</td></tr><tr><td>550</td><td>450</td><td>1230</td><td>600</td><td>3600</td></tr></table> Instructions: 1. Using available technology tools (e.g. MS Excel, Jamovi), input the data on the monthly electricity bills of 50 households. 2. Compute for the following measures of variability: a. Range b. Interquartile Range	1550	1450	1350	1500	1800	750	1550	1200	1300	1200	2500	1500	1600	1700	1500	2500	3000	3500	4000	3500	2300	3400	2300	2450	2000	2000	2100	1250	1300	1100	950	850	800	900	950	1250	1300	1200	1500	850	875	900	1350	750	780	550	450	1230	600	3600	Activity C.1: Watts Up? Investigating the Variations in Household Electricity Bills This activity is designed to be completed as group work. Ensure that all learners actively participate and collaborate effectively. This task is an opportunity for them to deepen their understanding and refine their skills, as it will help prepare them for the upcoming performance task. Encourage them to aim for thorough and high-quality work. Provide feedback on their output.
	1550	1450	1350	1500	1800																																															
	750	1550	1200	1300	1200																																															
	2500	1500	1600	1700	1500																																															
	2500	3000	3500	4000	3500																																															
	2300	3400	2300	2450	2000																																															
	2000	2100	1250	1300	1100																																															
	950	850	800	900	950																																															
	1250	1300	1200	1500	850																																															
	875	900	1350	750	780																																															
550	450	1230	600	3600																																																

- c. Variance
- d. Standard Deviation
- 3. Analyze and interpret the results.
- 4. Answer the following questions:
 - a. What patterns do you notice? Are household electricity bills clustered or spread out?
 - b. Which measures of variability best captures stability and instability in the household electricity bills?
 - c. How can variability inform better pricing structures for household electricity bills?

C.2. Making Generalization

Activity C.2: Wrapping-up the Lesson

Rock-Paper-Scissors Reflections

Instructions:

1. Learners will reflect on their learning using Rock-Paper-Scissors metaphor to organize their thoughts.
2. Learners will write their answer on the reflection table:
 - o Rock section- the most challenging part of the lesson (the concept that requires more effort to understand)
 - o Paper section - the main idea or the key takeaway from the lesson
 - o Scissors - less important details or extra information

Reflection Area	Answer	Explanation
ROCK		
PAPER		
SCISSORS		

2. Be specific and honest. Your answer will help your teacher understand what stood out to you.

Activity C.2: Wrapping-up the Lesson

For the summary, you may select and facilitate one of the provided activities.

Traffic Light Reimagining

Instructions:

1. Think about what you learned, what you are pondering upon, and what are you struggling today related to measures of variability.
2. Learners will reflect on their comfort level with each measure and classify their understanding using:

Red Light - Need Help

I struggle with this concept and need further explanation.

Yellow Light - Somewhat Confident

I understand the basics but need more practice.

Green Light – Confident

I can explain it to a peer and apply it to different real-life situations.

3. Pair up with a peer and compare your traffic light results.

4. Listen to how others connect the topic to their experiences.

Clear or Cloudy Exit Ticket

Instructions:

1. Learners will take a moment to complete the exit ticket:
 - **Clear** (one concept or skill from today's lesson that you understand well)
 - **Cloudy** (one concept or skill that is still confusing or unclear)
2. Learners will write their answers on a piece of paper using the following template:

Name:	
Clear	<i>(Something you fully understand)</i>
Cloudy	<i>(Something you need more practice or clarification on)</i>
Ways Forward	<i>(Step can you take to make your "cloudy" concept clearer)</i>

3. Be specific and honest. Your responses will help guide future discussions and review sessions.

C.3. Evaluating Learning

Activity C.3: Solving Real-life Problems Using Measures of Variability

Instructions:

Read and analyze the following problems and answer what is asked. Use available technology tools (e.g. MS Excel, Jamovi) in the computation process.

1. A teacher wants to determine and analyze the variability of daily transportation costs (in pesos) of learners since they travel to school using different modes of transportation. He/she collected the data from 30 learner-commuters to identify the patterns and trends. The data are given below.

55	43	105	34	80
25	32	60	55	90
34	36	45	41	47
50	55	56	59	49
29	30	45	57	59
40	40	45	50	55

Activity C.3: Solving Real-life Problems Using Measures of Variability

This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation.

- a. Compute for the range, interquartile range, variance, and standard deviation.
- b. Interpret the results.
- c. What measure of variability is the most suitable or appropriate for the given situation? Justify your answer.

2. A group of researchers surveyed 50 Grade 11 learners to analyze variability in learners' sleep duration (in hours). They divide learners into two groups (i.e. Group 1 – learners who study in the early evening and Group 2 – learners who study late at night. The following data were collected.

Group 1

5	6.5	9	8	7.5
6	6	7	7	8
5.5	5.5	6.5	7	5.5
4	5.5	7.5	6.5	8.5
8	4.5	5	7.5	9.5

Group 2

4	7	7	4.5	5.5
5	5.5	6	6.5	6
5	6	7	7.5	5.5
4.5	6	5	5.5	7.5
6	5	4.5	4.5	4.5

- a. For each group, compute for the range, interquartile range, variance, and standard deviation.
- b. Interpret the results.
- c. How does the variability in sleep duration compare between the groups? Which group appears to have more stable sleep durations? Which group has more extreme fluctuations in sleep hours?

C.4. Additional Activities

Activity C.4:

Enhancing the Skills on Measures of Variability Remediation

Instructions:

Answer the following problems:

1. A water utility company wants to analyze household water consumption variability to improve resource allocation and promote conservation. Below is the collected data of 20 households on their water consumption (in liters).

750	760	560	430
500	480	670	340
650	900	880	530
780	850	780	610
670	950	670	820

a. Calculate the following measures of variability.

- (i) range
 - (ii) interquartile range
 - (iii) variance
 - (iv) standard deviation
- b. Interpret the results.

2. Two groups of learners were asked about their mobile data consumption (i.e. Group 1 learners – mobile data usage for academic purposes and Group 2 learners – mobile data usage for social media purposes). The collected data on monthly mobile data consumption (in gigabytes) among 20 learners (10 for each group) are as follows.

Group 1		Group 2	
9	10.5	8	7.5
10	9.5	12	9

Activity C.4:

Enhancing the Skills on Measures of Variability

This additional exercise may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning.

8.5	7.5	13	10
9.5	13	16	10.5
12	11.5	6.5	11

a. For each group, calculate the following measures of variability.

(i) range

(ii) interquartile range

(iii) variance

(iv) standard deviation

b. Interpret the results.

c. Which group shows more consistent mobile data usage? Which group has more extreme fluctuations?

Enhancement:

Answer the following problem.

Gen and Mat own a small milktea shop and want to analyze the daily sales of three best-selling flavors: Black Milktea and Taro Milktea. They recorded the number of cups sold over 10 days:

<i>Black Milktea</i>	<i>Taro Milktea</i>	<i>Thai Milktea</i>
32	31	32
32	32	33
34	34	34
35	35	36
31	36	30
31	23	28
33	37	21
35	34	39
28	31	37
32	28	22

a. Find the range, interquartile range, variance, and standard deviation for each milktea flavor.

b. Interpret the computed measures of variability.

	c. Which flavor has the largest variability in sales? How about the smallest variability in sales? Explain your answer. d. How would adding data from another 10 days change the overall variability? e. What happens to the measures of variability (i) if we add 5 and (ii) if we multiply 2 to each observation in the given data set? f. How can understanding variability help them improve customer satisfaction?	
V. ASSESSMENTS	Please refer to the attached summative assessment.	
VI. REFLECTION	For the teachers, you may answer the following questions: 1. What went well in your teaching? 2. What are the challenges you encounter? 3. How did you manage these challenges?	

Summative Assessment

Instructions:

I. Choose the best answer that corresponds to the question.

1. The mode is the observation that occurs most frequently in a set. It is best used when dealing with what level of measurement?

- A. Nominal B. Ordinal C. Interval D. Ratio

2. The Learner Reference Number (LRN) is a unique and permanent 12-digit number assigned to anyone enrolled in the Philippine Basic Education System. It serves as a national student identification number in the Philippines. The LRN of a learner is an example of what kind of variables?

- A. qualitative, nominal C. quantitative, interval
B. qualitative, ordinal D. quantitative, ratio

3. If all the values of a data set are the same, which of the following measures will be equal to zero?

- A. range, variance, standard deviation, and interquartile range
B. mean, variance, standard deviation, and interquartile range
C. median, range, variance, and standard deviation
D. mean, range, variance, and standard deviation
E. mean, median, mode, variance, and standard deviation

4. Which of the following correctly matches a data type with its level of measurement?

- I. Social Security Number ▼ Nominal III. Temperature (°C) ▼ Interval
II. Place in oratorical contest ▼ Ordinal IV. Test scores out of 100 ▼ Ratio

- A. I, II, III B. I, II, IV C. II, III, IV D. I, II, III, IV

5. Which of the following statements is/are TRUE?

- I. Nominal data cannot be ranked in order of preference.
II. Ordinal data always has equal gaps between ranks.
III. Interval-level data can have negative values.
IV. Temperature in Celsius is considered ratio-level data.

- A. I only B. I and III C. I and IV D. II and IV

II. Analyze and solve the problems below.

1. A teacher asked his/her class how much time they spent studying for their General Mathematics examination. The results (in minutes) were as follows: {120, 75, 80, 240, 360, 150, 180, 50, 330, 125, 190, 95, 125, 380}.

- a. Compute for the (i) mean, (ii) median, and (iii) mode.
b. Which measures of central tendency would be the most appropriate measure to use in this situation? Why?
c. Calculate for the (i) range, (ii) interquartile range, (iii) variance, and (iv) standard deviation.
d. Interpret the result.

2. The table shows the average low and high temperature (°C) recorded in the Philippines for the year 2023.

Temp.	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
Low	23.2	23.2	24.0	25.1	26.4	26.1	25.7	25.7	25.3	25.2
High	28.2	29.5	30.9	32.8	33.3	31.9	30.7	30.2	30.6	30.6

Source: <https://www.weather-atlas.com/en/philippines/pag-asa-climate>

- Find for mean, median, and mode for low and high temperatures recorded throughout the year.
 - Which measure gives the best representation of typical low and high temperatures?
 - Determine the range, interquartile range, variance, and standard deviation of the given data.
 - Interpret the result.
3. During the past eight games in the basketball league in your town, the two teams recorded the following average points per game:

Team	Game 1	Game 2	Game 3	Game 4	Game 5	Game 6	Game 7	Game 8
Gen	82	75	77	80	75	79	78	72
Mat	72	70	65	82	85	89	85	69

- Find the mean, median, and mode for each team's score throughout the eight games played.
- Which measures best represent the performance of each team?
- Determine the range, interquartile range, variance, and standard deviation of the game performance for both teams.
- Which team has a wider spread in scores, and what does this indicate?

Performance Task

DDM (Data-Driven Decision Making) – How Central Tendency and Variability Shape Data

Act as data analysts solving real-world problems using measures of central tendency and variability. They will collect, analyze, and interpret data, draw conclusions, and propose practical solutions based on their findings.

Instructions:

- Choose one scenario for data collection. Sample scenarios are given below.
 - Classroom Performance
Analyze quiz scores from classmates to determine overall class performance.
 - Family Spending
Examine weekly grocery expenses to assess budgeting patterns.
 - Screen Time Analysis
Compare daily phone usage of learners to identify trends in digital habits.
 - Weather Patterns
Record daily temperatures over two weeks to study climate trends.

Note: You can also create their own real-world scenario.

- For data gathering, collect at least 20-25 data from sources (e.g. class surveys, electronic databases, classroom observations). Organize the data in a table.
- Compute for the following measures:
 - central tendency (mean, median, mode)
 - variability (range, interquartile range, variance, standard deviation)
- Answer the following questions:
 - What does the central tendency reveal about typical patterns in your dataset?
 - Which variability measure helped you understand consistency or instability best?

- c. What real-life recommendations can be made based on the findings?
5. For presentation of the findings, learners may choose one from the following format:
- Infographic**
Create a poster or an infographic summarizing the data trends. You may use Canva, Piktochart, or PowerPoint.
 - News Report**
Record a video or perform a live class presentation in the style of a news update.
 - Storytelling or Comic Strip**
Present the data in the form of a creative story or comic strip.
 - Gamified Presentation**
Present findings through an interactive quiz, scavenger hunt, or puzzle game.
 - Podcast**
Record a discussion where learners explain findings conversationally.
 - Short Film or Skit**
Create a mini movie showcasing the data investigation process.
 - Debate**
Defend the best measure of central tendency or variability for their scenario.
6. For reflection questions, learners will answer the following:
- How did measures of central tendency help summarize the dataset?
 - Which measures of variability were most useful for analyzing real-world patterns?
 - If you collected more data, how might your findings change?
 - Explain how understanding variability and central tendency helps in decision-making.

Assessment Rubric (50 points)

Criteria	4 - Exemplary <i>(10 points)</i>	3 - Proficient <i>(8 points)</i>	2 - Developing <i>(6 points)</i>	1 - Beginning <i>(4 points)</i>
Data Collection and Organization 20%	Well-organized, accurate data	Mostly accurate and organize, few inconsistencies	Limited data or missing values	Little to no data collected
Accuracy of Computations 35%	All measures are computed correctly and accurately.	Most measures are computed accurately with minor errors	Some errors in key calculations	Multiple major errors in computations

Analysis and Interpretation 20%	Written analysis and interpretation are concise, well-structured, and free from grammar or structure issues	Analysis and interpretation are mostly clear; minor grammar or structure issues	Analysis and interpretation are unclear or has many writing issues	No clear analysis and interpretation
Creativity and Quality of Presentation 10%	Well-structured, engaging, and appealing visuals and report and presentation effectively conveys message	Clear visuals with minor formatting issues and moderately engaging presentation	Basic structure or have limited visuals and limited effort in presentation	Poorly structured and minimal or missing presentation
Real-World Application 15%	Real-world connections are thoughtful, practical, and well-supported	Some connection to practical applications but lacks depth	Weak relevance to real-world usage or unclear impact	No application of findings